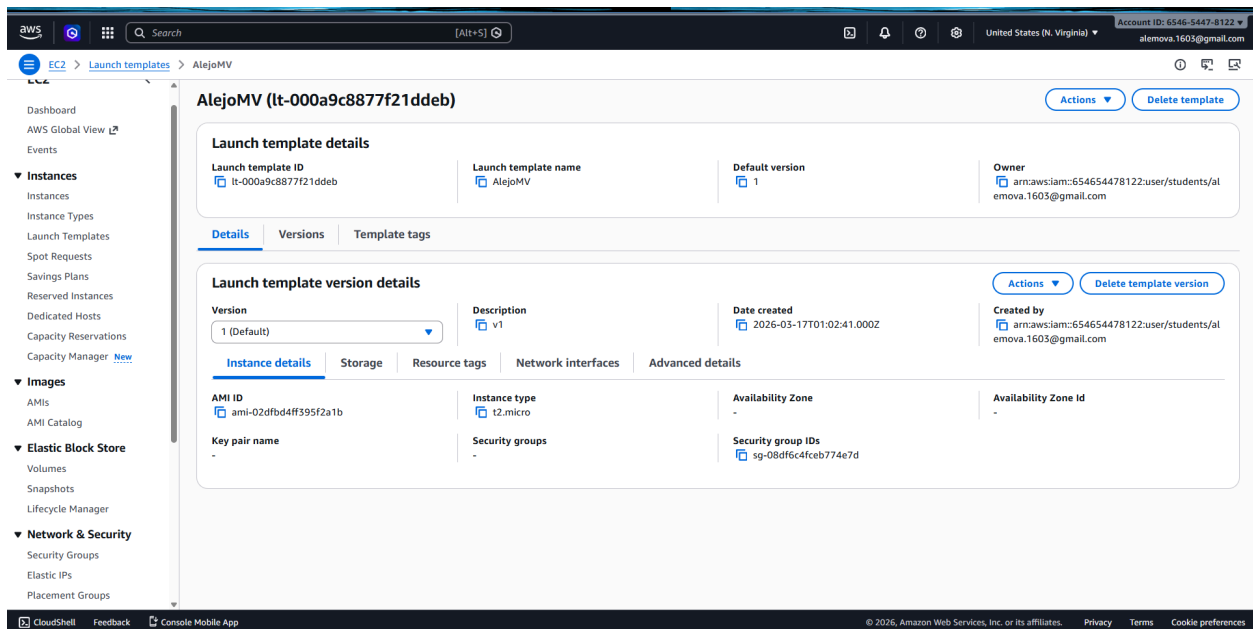


Evidencia evaluación practica - Modulo 2

Alejandro Montoya Vargas - alemova.1603@gmail.com

Parte 1: Configurar la Escalabilidad (Auto Scaling)

1. Launch Template: Crear una plantilla de lanzamiento (Launch Template) con una instancia t2.micro (Amazon Linux 2023).



2. Auto Scaling Group (ASG): * Crear un grupo que use el template anterior.
 - o Configurar capacidad: Deseada: 1, Mínima: 1, Máxima: 2
- 3.

AlejoMV-AutoScalingGroup Capacity overview

arm:aws:autoscaling:us-east-1:654654478122:autoScalingGroup:7789e03c-c66f-4b11-8fef-61e0e4d36907:autoScalingGroupName/AlejoMV-AutoScalingGroup

Desired capacity	Scaling limits	Desired capacity type	Status
1	1 - 2	Units (number of instances)	-

Date created
Mon Mar 16 2026 20:10:56 GMT-0500 (hora estándar de Colombia)

Details | **Integrations** | **Automatic scaling** | **Instance management** | **Instance refresh** | **Activity** | **Monitoring** | **Tags**

Instances (1)

Filter instances

Instance ID	Lifecycle	Instance type	Weighted capacity	Launch template...	Availability Zone	Health status	Protected from
i-081191f374db88f5c	InService	t2.micro	-	AlejoMV Version 1	us-east-1a (us-east-1a)	Healthy	

Instance lifecycle policy for lifecycle hooks

Controls instance behavior when an instance transitions through its lifecycle states.

Termination hook abandon behavior
Terminate (default)

Lifecycle hooks (0)

Instance summary for i-081191f374db88f5c (AlejoMv)

Updated less than a minute ago

Instance ID	Public IPv4 address	Private IPv4 addresses
i-081191f374db88f5c	44.193.200.241 open address	172.31.7.218

Instance state
Running

Hostnames
IP name: ip-172-31-7-218.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-7-218.ec2.internal

Instance type
t2.micro

VPC ID
vpc-00f479057476a2db8 (VPC-Produccion)

Subnet ID
subnet-00227b47a5f26ec04

Instance ARN
arn:aws:ec2:us-east-1:654654478122:instance/i-081191f374db88f5c

IAM role
-

IMDSv2
Required

Operator
-

Auto-assigned IP address
44.193.200.241 [Public IP]

Public DNS
ec2-44-193-200-241.compute-1.amazonaws.com | open address

Elastic IP addresses
-

AWS Compute Optimizer finding
User: arn:aws:iam::654654478122:user/students/alemovia.1603@gmail.com is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action
Retry

Auto Scaling Group name
AlejoMV-AutoScalingGroup

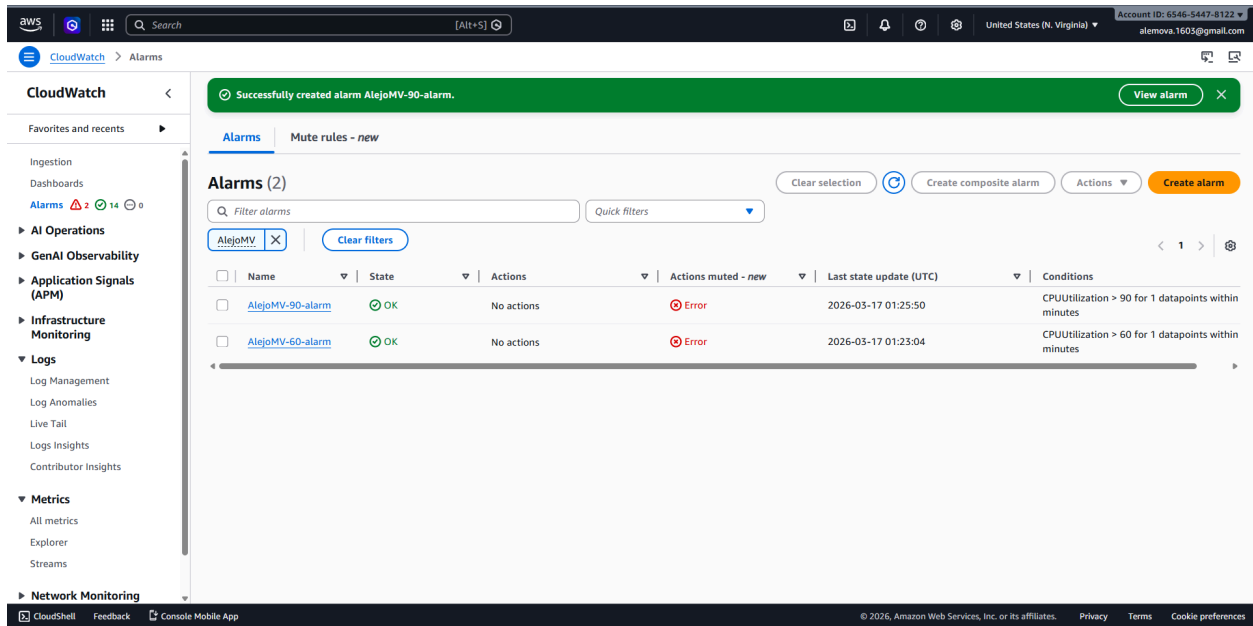
Managed
false

Details | **Status and alarms** | **Monitoring** | **Security** | **Networking** | **Storage** | **Tags**

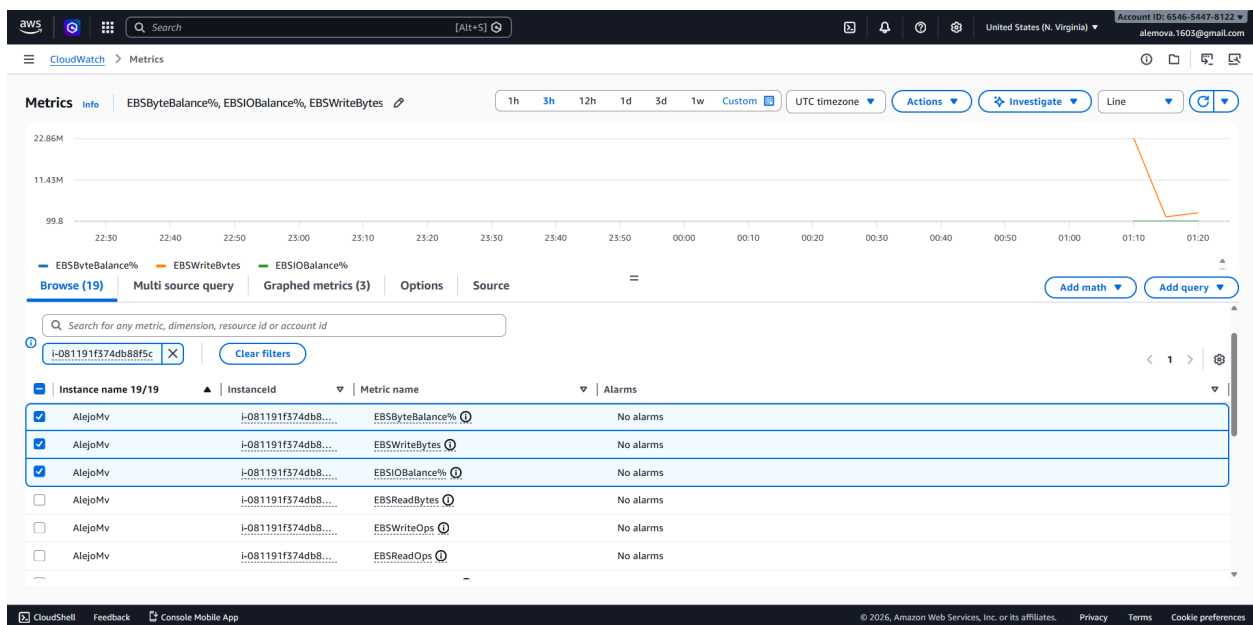
Instance details

Parte 2: Crear una alarma en CloudWatch

1. Crea dos alarmas de CPU de la instancia que creó el auto scaling Group, una que sea sobre el 60% de CPU de advertencia y otra del 90% de CPU de alerta roja

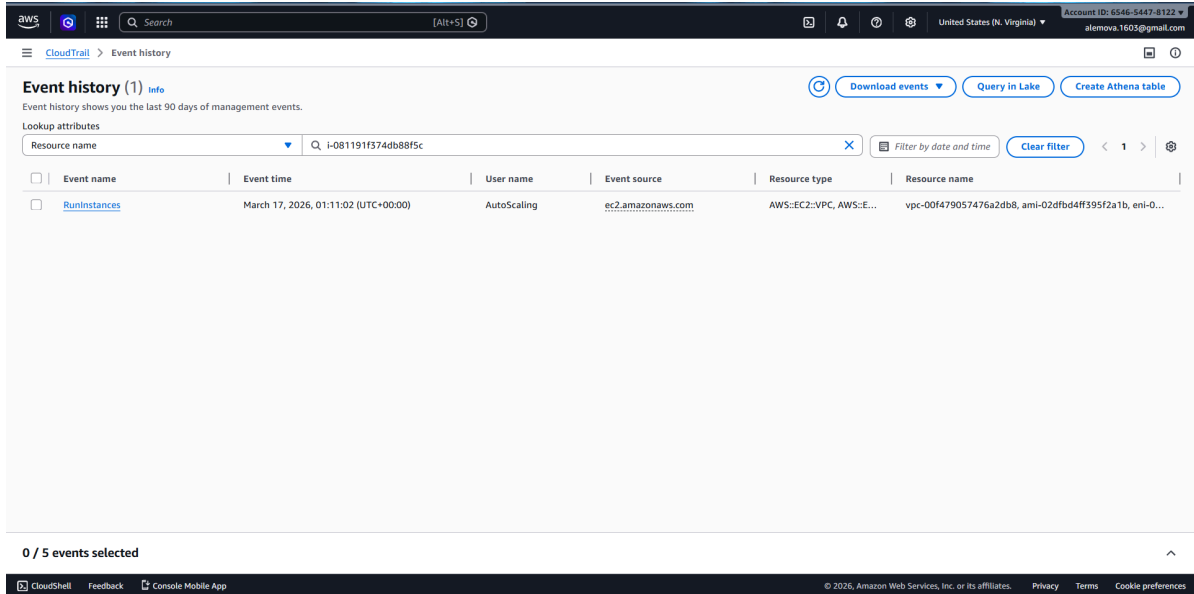


2. Revisa desde la consola de cloudwatch todas las métricas asociadas a la instancia
coloca por lo menos 3 donde se vea el id de la instancia.



Parte 3: Revisar eventos en CloudTrail

1. En cloud Trail ingresa y busca el evento de creación de una de las instancias, pone el JSON e identifica desde donde se creó la instancia.



Se puede ver desde donde fue creado y por quien:

- "invokedBy": "autoscaling.amazonaws.com"
- "sourceIPAddress": "autoscaling.amazonaws.com"
- Y el específico:

```

    "tagSpecificationSet": {
      "items": [
        {
          "resourceType": "instance",
          "tags": [
            {
              "key": "aws:autoscaling:groupName",
              "value": "AlejoMV-AutoScalingGroup"
            },
            {
              "key": "Name",
              "value": "AlejoMv"
            }
          ]
        }
      ]
    },
    "launchTemplate": {

```

[Skip to Main Content](#)

Search
[Alt+S]

United States (N. Virginia)

Account ID: 6546-5447-8122
alemova.1603@gmail.com

CloudTrail
Event history
RunInstances
CloudTrail

Event history
RunInstances

Event details - CloudTrail

RunInstances Info

Details Info

Event time

March 17, 2026, 01:11:02 (UTC+00:00)

User name

AutoScaling

Event name

RunInstances

Event source

ec2.amazonaws.com

AWS access key

ASIAZQ3DSS4VFMJB7MMA

Source IP address

autoscaling.amazonaws.com

Event ID

6fb9426c-24fd-4db3-ac8f-30b942fe5418

Request ID

8e6e286c-d23e-4817-b8d7-04a80d0d6ef1

AWS region

us-east-1

Error code

-

Read-only

false

Resources referenced (7) Info

Resources referenced describes the name or ID of resources that were read or changed by an event

Resource type

Resource name

AWS Config resource timeline

AWS::EC2::VPC

vpc-00f479057476a2db8

Enable AWS Config resource recording

AWS::EC2::Ami

ami-02dfbd4ff395f2a1b

Enable AWS Config resource recording

AWS::EC2::NetworkInterface

eni-0c429240cf08893c6

Enable AWS Config resource recording

AWS::EC2::Instance

i-081191f374db88f5c

Enable AWS Config resource recording

AWS::EC2::SecurityGroup

sg-08df6c4fceb774e7d

Enable AWS Config resource recording

AWS::EC2::SecurityGroup

AlejoMV-EC2

Enable AWS Config resource recording

AWS::EC2::Subnet

subnet-00227b47a5f26ec04

Enable AWS Config resource recording

Event record Info

Copy

JSON view

```
{
  "eventVersion": "1.11",
  "userIdentity": {
    "type": "AssumedRole",
    "principalId": "AR0AZQ3DSS4VILINXU7ZK:AutoScaling",
    "arn": "arn:aws:sts::654654478122:assumed-role/AWSServiceRoleForAutoScaling/AutoScaling",
    "accountId": "654654478122",
    "accessKeyId": "ASIAZQ3DSS4VFMJB7MMA",
    "sessionContext": {
```

```

        "sessionIssuer": {
            "type": "Role",
            "principalId": "AR0AZQ3DSS4VILINXU7ZK",
            "arn": "arn:aws:iam::654654478122:role/aws-se
service-role/autoscaling.amazonaws.com/AWSServiceRoleForAutoSca
ling",
            "accountId": "654654478122",
            "userName": "AWSServiceRoleForAutoScaling"
        },
        "attributes": {
            "creationDate": "2026-03-17T01:01:01Z",
            "mfaAuthenticated": "false"
        }
    },
    "invokedBy": "autoscaling.amazonaws.com"
},
"eventTime": "2026-03-17T01:11:02Z",
"eventSource": "ec2.amazonaws.com",
"eventName": "RunInstances",
"awsRegion": "us-east-1",
"sourceIPAddress": "autoscaling.amazonaws.com",
"userAgent": "autoscaling.amazonaws.com",
"requestParameters": {
    "instancesSet": {
        "items": [
            {
                "minCount": 1,
                "maxCount": 1
            }
        ]
    }
},
"blockDeviceMapping": {},
"availabilityZone": "us-east-1a",
"monitoring": {
    "enabled": false
},

```



```

    "disableApiTermination": false,
    "disableApiStop": false,
    "clientToken": "5b4673e5-6283-0263-24b3-3c5c5b0bfea
3",
    "networkInterfaceSet": {
      "items": [
        {
          "deviceIndex": 0,
          "subnetId": "subnet-00227b47a5f26ec04"
        }
      ]
    },
    "tagSpecificationSet": {
      "items": [
        {
          "resourceType": "instance",
          "tags": [
            {
              "key": "aws:autoscaling:groupNam
e",
              "value": "AlejoMV-AutoScalingGrou
p"
            },
            {
              "key": "Name",
              "value": "AlejoMv"
            }
          ]
        }
      ]
    },
    "launchTemplate": {
      "launchTemplateId": "lt-000a9c8877f21ddeb",
      "version": "1"
    }
  },

```

```

"responseElements": {
  "requestId": "8e6e286c-d23e-4817-b8d7-04a80d0d6ef1",
  "reservationId": "r-01ae22db3c70c3d72",
  "ownerId": "654654478122",
  "groupSet": {},
  "instancesSet": {
    "items": [
      {
        "instanceId": "i-081191f374db88f5c",
        "imageId": "ami-02dfbd4ff395f2a1b",
        "bootMode": "uefi-preferred",
        "currentInstanceBootMode": "legacy-bios",
        "instanceState": {
          "code": 0,
          "name": "pending"
        },
        "privateDnsName": "ip-172-31-7-218.ec2.in
ternal",
        "operator": {
          "managed": false
        },
        "amiLaunchIndex": 0,
        "productCodes": {},
        "instanceType": "t2.micro",
        "launchTime": 1773709862000,
        "placement": {
          "availabilityZone": "us-east-1a",
          "availabilityZoneId": "use1-az1",
          "tenancy": "default"
        },
        "monitoring": {
          "state": "disabled"
        },
        "subnetId": "subnet-00227b47a5f26ec04",
        "vpcId": "vpc-00f479057476a2db8",
        "privateIpAddress": "172.31.7.218",

```

```

    "stateReason": {
      "code": "pending",
      "message": "pending"
    },
    "architecture": "x86_64",
    "rootDeviceType": "ebs",
    "rootDeviceName": "/dev/xvda",
    "blockDeviceMapping": {},
    "virtualizationType": "hvm",
    "hypervisor": "xen",
    "tagSet": {
      "items": [
        {
          "key": "aws:ec2launchtemplat
e:id",
          "value": "lt-000a9c8877f21dde
b"
        },
        {
          "key": "aws:ec2launchtemplat
e:version",
          "value": "1"
        },
        {
          "key": "Name",
          "value": "AlejoMv"
        },
        {
          "key": "aws:autoscaling:group
Name",
          "value": "AlejoMV-AutoScaling
Group"
        }
      ]
    },
    "clientToken": "5b4673e5-6283-0263-24b3-3

```

```

c5c5b0bfea3",
    "groupSet": {
      "items": [
        {
          "groupId": "sg-08df6c4fceb774
e7d",
          "groupName": "AlejoMV-EC2"
        }
      ]
    },
    "sourceDestCheck": true,
    "networkInterfaceSet": {
      "items": [
        {
          "networkInterfaceId": "eni-0c
429240cf08893c6",
          "subnetId": "subnet-00227b47a
5f26ec04",
          "vpcId": "vpc-00f479057476a2d
b8",
          "ownerId": "654654478122",
          "operator": {
            "managed": false
          },
          "status": "in-use",
          "macAddress": "02:bd:ae:aa:1
2:2d",
          "privateIpAddress": "172.31.
7.218",
          "privateDnsName": "ip-172-31-
7-218.ec2.internal",
          "sourceDestCheck": true,
          "interfaceType": "interface",
          "groupSet": {
            "items": [
              {

```

```

        "groupId": "sg-08
df6c4fceb774e7d",
        "groupName": "Ale
joMV-EC2"
    }
]
},
"attachment": {
    "attachmentId": "eni-atta
ch-0bd6ae99e9a978332",
    "deviceIndex": 0,
    "networkCardIndex": 0,
    "status": "attaching",
    "attachTime": 17737098620
    "deleteOnTermination": tr
ue
},
"privateIpAddressesSet": {
    "item": [
        {
            "privateIpAddres
s": "172.31.7.218",
            "privateDnsName":
"ip-172-31-7-218.ec2.internal",
            "primary": true
        }
    ]
},
"ipv6AddressesSet": {},
"tagSet": {}
}
]
},
"ebsOptimized": false,
"enaSupport": true,

```

```

        "cpuOptions": {
            "coreCount": 1,
            "threadsPerCore": 1
        },
        "capacityReservationSpecification": {
            "capacityReservationPreference": "open"
        },
        "enclaveOptions": {
            "enabled": false
        },
        "metadataOptions": {
            "state": "pending",
            "httpTokens": "required",
            "httpPutResponseHopLimit": 2,
            "httpEndpoint": "enabled",
            "httpProtocolIpv4": "enabled",
            "httpProtocolIpv6": "disabled",
            "instanceMetadataTags": "disabled"
        },
        "maintenanceOptions": {
            "autoRecovery": "default",
            "rebootMigration": "default"
        },
        "privateDnsNameOptions": {
            "hostnameType": "ip-name",
            "enableResourceNameDnsARecord": false,
            "enableResourceNameDnsAAAARecord": false
        }
    ],
    "requesterId": "661627717883"
},

```

```
"requestID": "8e6e286c-d23e-4817-b8d7-04a80d0d6ef1",
"eventID": "6fb9426c-24fd-4db3-ac8f-30b942fe5418",
"readOnly": false,
"resources": [
  {
    "accountId": "654654478122",
    "type": "AWS::EC2::Instance",
    "ARN": "arn:aws:ec2:us-east-1:654654478122:instance/i-081191f374db88f5c"
  }
],
"eventType": "AwsApiCall",
"managementEvent": true,
"recipientAccountId": "654654478122",
"eventCategory": "Management"
}
```

[CloudShell](#)

[Feedback](#)

[Console Mobile App](#)

© 2026, Amazon Web Services, Inc. or its affiliates.

[Privacy](#)

[Terms](#)

[Cookie preferences](#)

2. Revisa qué acciones ha realizado tu usuario de iam y pon las últimas 5 acciones, sin detalle.

Event history (50+) [Info](#)

Event history shows you the last 90 days of management events.

Lookup attributes

User name: alemova.1603@gmail.com [Filter by date and time](#) [Clear filter](#)

☐	Event name	Event time	User name	Event source	Resource type	Resource name
☐	LookupEvents	March 17, 2026, 01:40:55 (UTC+00:00)	alemova.1603@g...	cloudtrail.amazonaws.com	-	-
☐	LookupEvents	March 17, 2026, 01:40:55 (UTC+00:00)	alemova.1603@g...	cloudtrail.amazonaws.com	-	-
☐	DescribeConfigurationRecorderS...	March 17, 2026, 01:40:29 (UTC+00:00)	alemova.1603@g...	config.amazonaws.com	-	-
☐	LookupEvents	March 17, 2026, 01:40:29 (UTC+00:00)	alemova.1603@g...	cloudtrail.amazonaws.com	-	-
☐	LookupEvents	March 17, 2026, 01:40:29 (UTC+00:00)	alemova.1603@g...	cloudtrail.amazonaws.com	-	-
☐	DescribeConfigurationRecorderS...	March 17, 2026, 01:35:12 (UTC+00:00)	alemova.1603@g...	config.amazonaws.com	-	-
☐	ListManagedNotificationEvents	March 17, 2026, 01:34:49 (UTC+00:00)	alemova.1603@g...	notifications.amazonaws.com	-	-
☐	LookupEvents	March 17, 2026, 01:34:22 (UTC+00:00)	alemova.1603@g...	cloudtrail.amazonaws.com	-	-
☐	LookupEvents	March 17, 2026, 01:34:21 (UTC+00:00)	alemova.1603@g...	cloudtrail.amazonaws.com	-	-
☐	GetAccountPlanState	March 17, 2026, 01:32:49 (UTC+00:00)	alemova.1603@g...	freetier.amazonaws.com	-	-
☐	DescribeRegions	March 17, 2026, 01:32:49 (UTC+00:00)	alemova.1603@g...	ec2.amazonaws.com	-	-
☐	GetAccountInformation	March 17, 2026, 01:32:49 (UTC+00:00)	alemova.1603@g...	account.amazonaws.com	-	-
☐	ListManagedNotificationEvents	March 17, 2026, 01:32:49 (UTC+00:00)	alemova.1603@g...	notifications.amazonaws.com	-	-

0 / 5 events selected

Parte 4 : Creación de función lambda

1. En esta actividad deberá crear una función AWS Lambda que consulte las instancias EC2

Primero creamos el rol:

Identity and Access Management (IAM)

AlejoMV-modulo2-lambdaRole [Info](#)

Allows Lambda functions to call AWS services on your behalf.

Summary

Creation date: March 16, 2026, 20:48 (UTC-05:00)

Last activity: 9 minutes ago

ARN: arn:aws:iam::654654478122:role/AlejoMV-modulo2-lambdaRole

Maximum session duration: 1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (2) [Info](#)

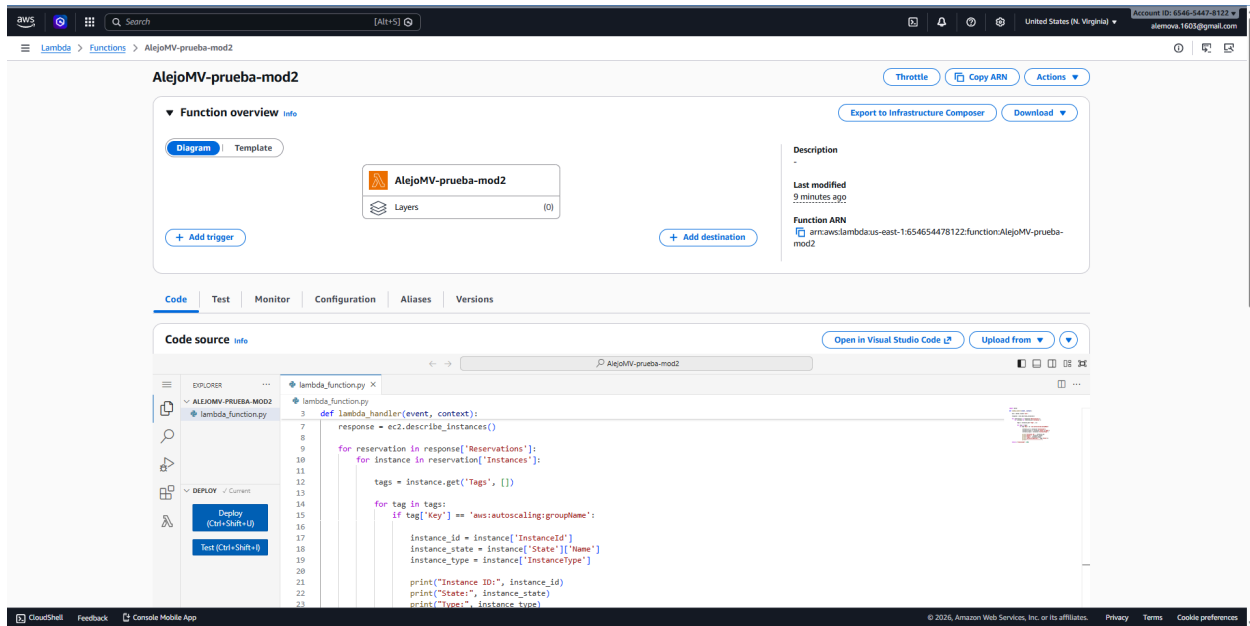
You can attach up to 10 managed policies.

☐	Policy name	Type	Attached entities
☐	AmazonEC2ReadOnlyAccess	AWS managed	10
☐	CloudWatchLogsFullAccess	AWS managed	4

Permissions boundary (not set)

Generate policy based on CloudTrail events

Luego la lambda de python:



Código:

```
import boto3

def lambda_handler(event, context):

    ec2 = boto3.client('ec2')

    response = ec2.describe_instances()

    for reservation in response['Reservations']:
        for instance in reservation['Instances']:

            tags = instance.get('Tags', [])

            for tag in tags:
                if tag['Key'] == 'aws:autoscaling:groupName':

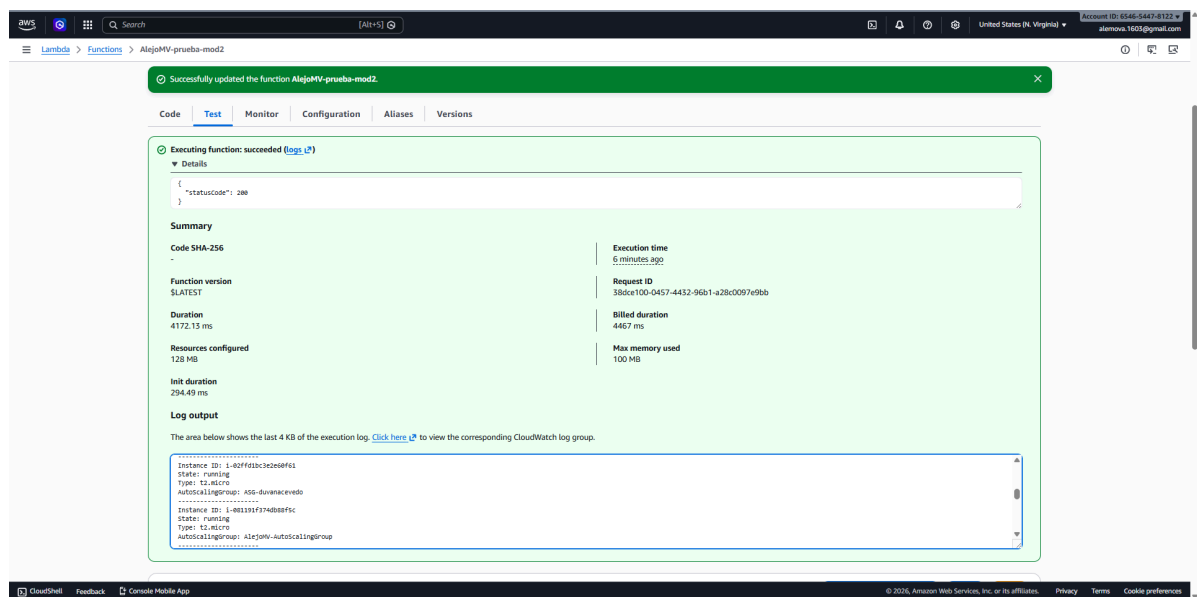
                    instance_id = instance['InstanceId']
                    instance_state = instance['State']['Name']
                    instance_type = instance['InstanceType']
```

```
print("Instance ID:", instance_id)
print("State:", instance_state)
print("Type:", instance_type)
print("AutoScalingGroup:", tag['Value'])
print("-----")
```

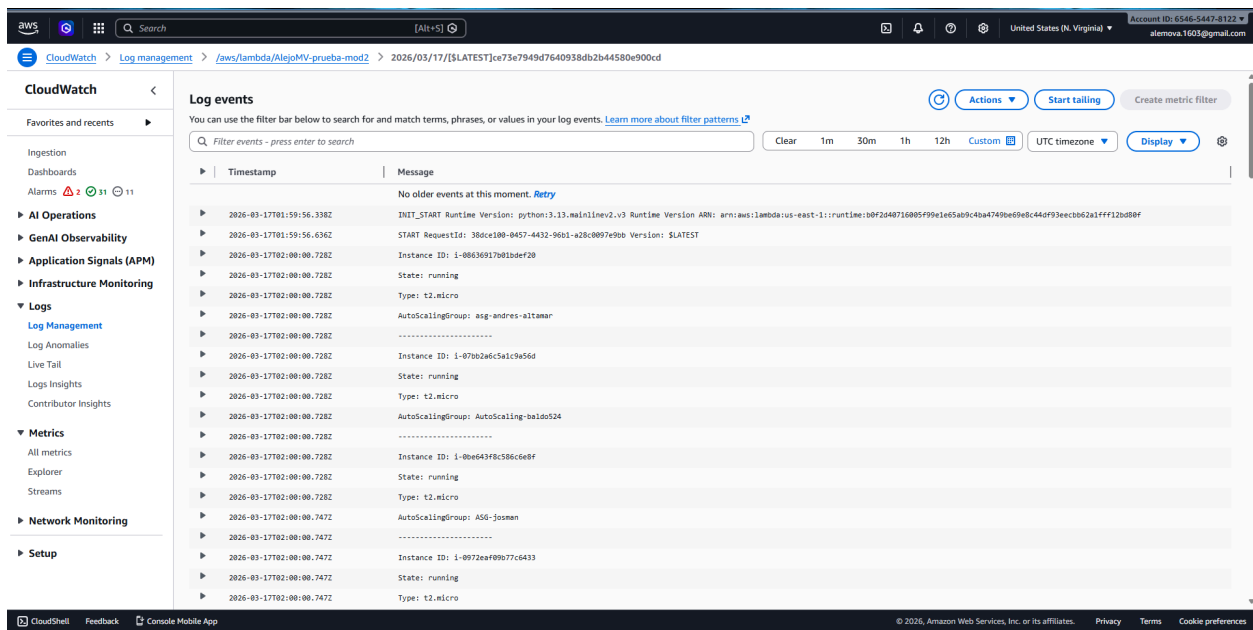
```
return {"statusCode": 200}
```

2. Muestra la siguiente información de cada una.

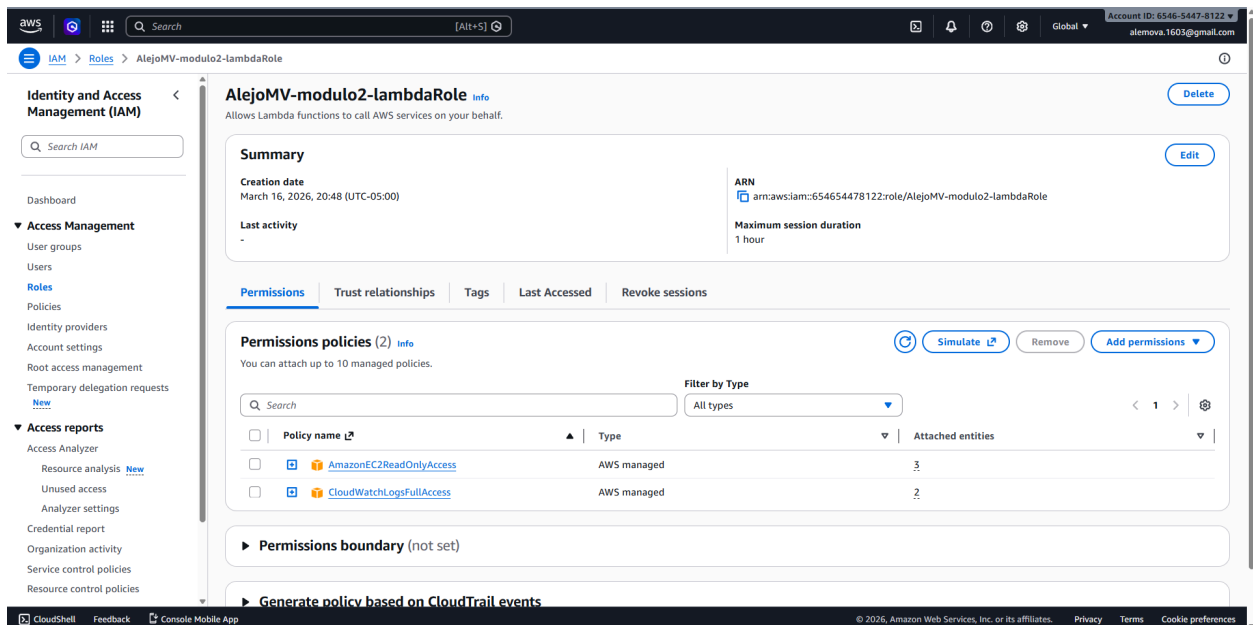
- ID de la instancia
- Estado
- Tipo de instancia



Aquí están todas:



3. Crear un IAM Role para la función Lambda



4. Verificar en CloudWatch Logs que la función haya registrado correctamente la información de las instancias, puntualmente el log de la lambda que implementó.

Aquí esta el log de mi lambda en específico.

AI Operations	2026-03-17T02:00:00.767Z	-----
GenAI Observability	2026-03-17T02:00:00.767Z	Instance ID: i-081191f374d088f5c
Application Signals (APM)	Instance ID: i-081191f374d088f5c	
Infrastructure Monitoring	2026-03-17T02:00:00.767Z	State: running
Logs	State: running	
Log Management		
Log Anomalies	2026-03-17T02:00:00.767Z	Type: t2.micro
Live Tail	Type: t2.micro	
Logs Insights		
Contributor Insights	2026-03-17T02:00:00.767Z	AutoScalingGroup: A1ej0W-AutoScalingGroup
Metrics	AutoScalingGroup: A1ej0W-AutoScalingGroup	